

Lily Lab: Take a Flower Apart, Then Keep It

Name: _____ Group: _____ Date: _____

**I can take a lily apart, name its reproductive parts and explain each part's job in making seeds.
I can press and mount those parts as a labelled specimen that will last for years.**

DAY 1 • ONE CLASS

Predict, dissect, record, and set up the press.

DAYS 2 & 4 • 5 MIN

Swap the wet blotter paper for dry. Check for mould.

~2 WEEKS LATER • ONE CLASS

Mount the dry parts on archival card and label them.

Predict Who Is the Lily For?

A lily is big, brightly coloured, spotted, and strongly scented. Predict: **who or what** is all of that for? Explain your thinking.

Safety Before You Touch the Lily

⚠ LAB RULES

- **Pollen stains.** Lily pollen stains skin, clothes, and paper orange. Don't rub it. Lift it off with tape.
- **Allergies?** Tell your teacher before you start. Gloves are available.
- **Don't eat any part** of the lily.
- **Lilies are poisonous to cats**, even the pollen. If you have a cat, don't take any lily parts home.
- **Cut away from your hand.** Only your teacher cuts the ovary.
- **Wash your hands** when you finish.

🧰 YOUR GROUP NEEDS

- 1 fully open lily + 1 lily leaf
- tray or paper plate
- small scissors, hand lens
- clear tape, paper towel
- 2 sheets **acid-free archival paper**
- 4 sheets blotter paper or newsprint
- 2 pieces of cardboard
- pencil (not pen) for labels

Roles Group Jobs

Dissector

Removes each part and lays it in the right tray column.

Name: _____

Pollen Keeper

Takes the tape sample. Keeps pollen off hands and paper.

Name: _____

Checker

Reads each step aloud. Checks every part is counted.

Name: _____

Press Manager

Builds and labels the press. Swaps blotters on Days 2 & 4.

Name: _____

Group of 3? The Checker is also the Pollen Keeper. **Everyone removes at least one part** and everyone fills in their own sheet.

From memory What Do You Already Know?

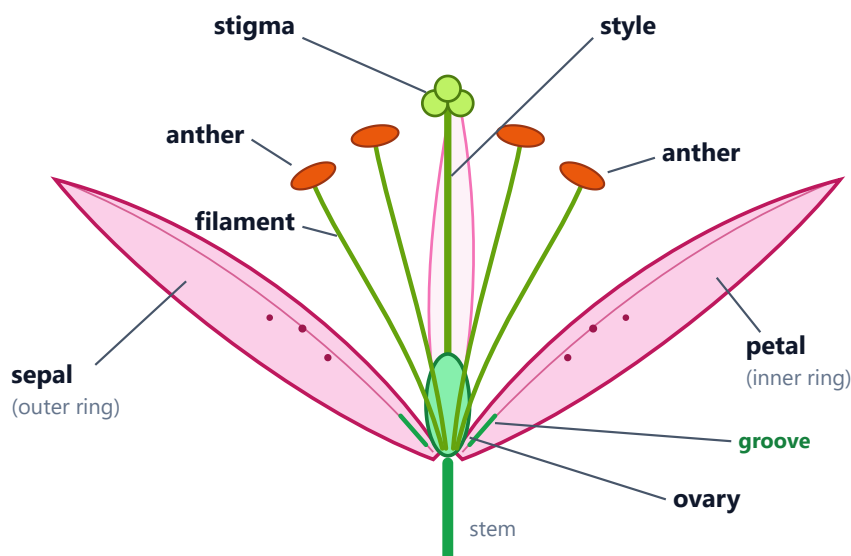
Don't look at your notes. Write the job of each flower part from memory. After the dissection, check your answers and fix any mistakes in the last column.

Part	Its job (from memory)	After the lab: ✓ correct, or the fix
Sepal		
Petal		
Stamen		
Pistil		

Read The Lily Trick

On most flowers the **sepals** are small and green. On a lily, the 3 sepals look almost exactly like the 3 petals. Botanists call all 6 of them **tepals**. You can still tell them apart. The **sepals are the outer ring**: they attach a little lower and further out, and they wrapped the outside of the bud. The **petals are the inner ring**: they are usually a little wider. All 6 tepals have a green **nectar groove** at the base.

Model The Parts of a Lily

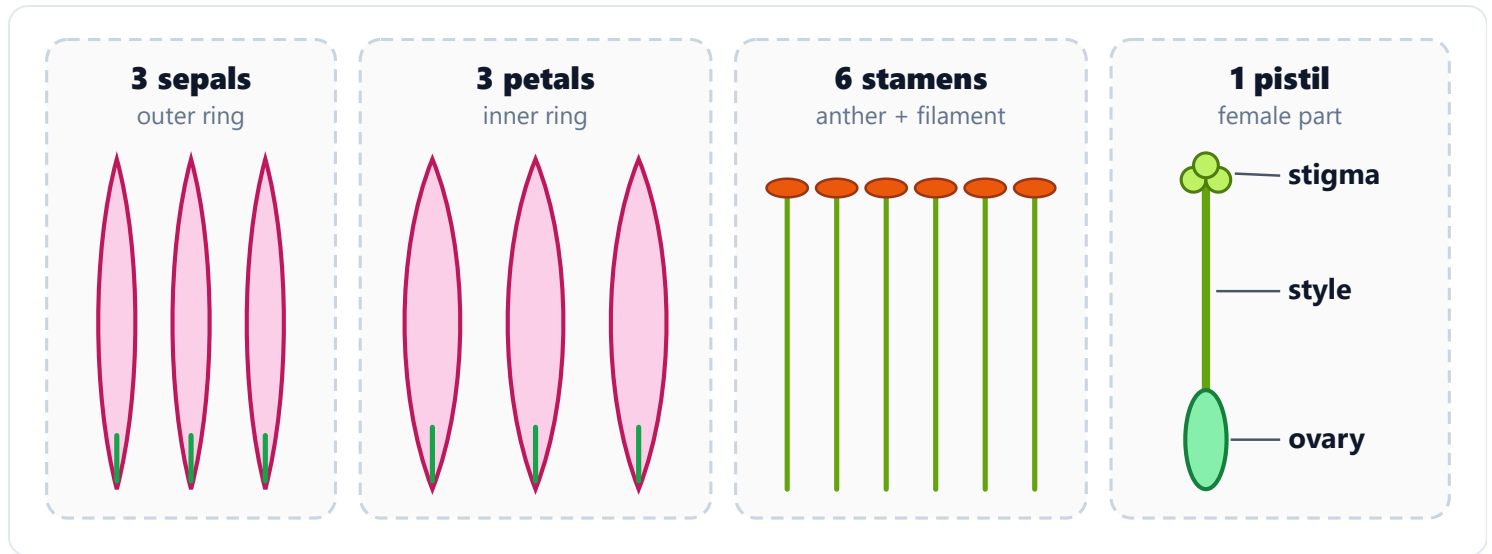


Stamen = anther + filament. The male part. The anther makes **pollen**. A lily has 6.

Pistil = stigma + style + ovary. The female part. The sticky stigma catches pollen. The ovary holds **ovules**, which can become **seeds**.

Part A Dissect: Outside In

Work from the outside of the flower to the centre. Lay each part on your tray in columns, like this model. Your tray should look like this when you finish.



Steps Procedure

1. **Look before you cut.** Count the tepals and stamens. Find the pistil in the centre. Smell the flower.
2. **Pollen sample.** Pollen Keeper: press a short piece of clear tape onto one anther and lift it off. Stick the tape in the pollen box on the next page.
3. **Sepals.** Find the 3 tepals in the outer ring. Pull each one gently down and off at its base. Find the green nectar groove at the base. Lay them in column 1.
4. **Petals.** Remove the 3 inner tepals. Compare their width to the sepals. Lay them in column 2.
5. **Stamens.** Pinch each filament at its base and pull. Hold it by the filament, not the anther. Lay all 6 in column 3.
6. **Pistil.** Only the pistil is left. Snip it off at the very bottom of the ovary. Lay it in column 4.
7. **Hand lens.** Look closely at the stigma and one anther. Record on the next page.
8. **Teacher station.** Bring your pistil to your teacher, who will cut a thin slice across the ovary. Look at the slice with the hand lens and sketch it.
9. **Keep everything.** Don't throw anything away. You will press these parts in Part B.

Record

What We Found

Part	How many?	Where on the flower?	What I observed (colour, texture, size, smell)	Its job in making seeds
Sepals				
Petals				
Stamens				
Pistil				

Pollen sample

What colour is it? What does it look like under the hand lens?

stick your tape here

Stigma close-up

Sketch the tip of the stigma. Touch it gently with a clean pencil tip.

It feels: _____

Ovary slice

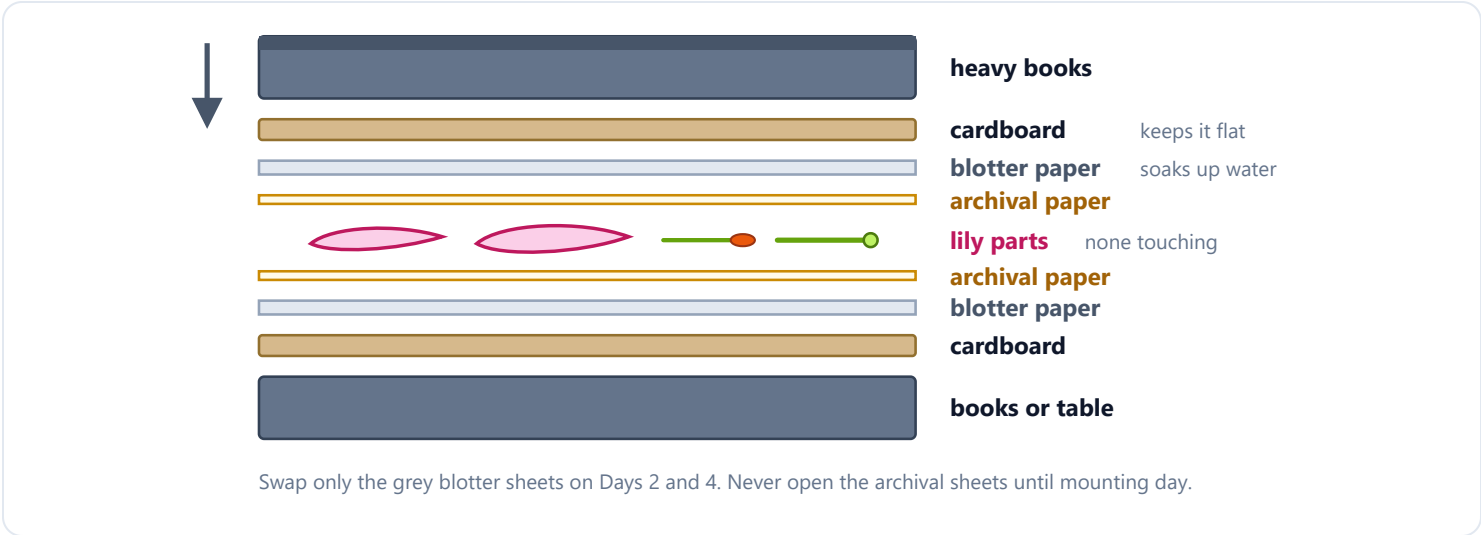
Sketch the slice. Label a **chamber** and an **ovule**.

Chambers: _____ Ovules I can count: _____

- A lily's sepals look almost the same as its petals. Explain how your group decided which tepals were sepals. What evidence did you use?
- Describe how the stigma felt. Explain how that texture helps the stigma do its job.
- Every ovule could become one seed. Use your ovule count to estimate how many seeds this one flower could make. What has to happen first before any ovule can become a seed?

Part B Press: The Archival Sandwich

Pressing flattens each part and pulls the water out of it slowly, so it dries flat instead of shrivelling. The parts sit between two sheets of **acid-free archival paper**. Blotter paper on the outside soaks up the water.



Steps Build Your Press

1. **Choose what to press:** 1 sepal, 1 petal, 2 stamens, the stigma and style (snip off the ovary), your teacher's ovary slice, and the leaf. Keep the rest as spares.
2. **Tap the anthers** over scrap paper to shake off loose pollen. Loose pollen stains the archival paper.
3. Lay down cardboard, then blotter, then one archival sheet. **Spread the parts out** so none touch or overlap. Smooth petals flat with a clean finger.
4. Cover with the second archival sheet, then blotter, then cardboard.
5. Write your group name **in pencil** on the top cardboard. Stack heavy books on top in a dry spot.
4. Predict: which part will press best, and which will be hardest to press? Explain using what you noticed about each part's **thickness and how much water** is inside it.

5. Why do you need to swap the wet blotter paper after 2 days instead of leaving it for the whole two weeks?

Press log	Date	What the parts look like (colour, still damp?, any mould?)
Day 2		
Day 4		

Part C Mount and Label (About 2 Weeks Later)

Your parts are ready when they feel dry and papery, not cool or bendy. Scientists store pressed plants like this in a **herbarium**, a library of plant specimens. Some are over 200 years old.

1. **Plan first.** Sketch your layout in the small sheet on the right. Put the parts in order, outside to inside, like your tray.
2. **Handle by the edges.** Dry parts crack. Lift them with a flat paper strip, not your fingers.
3. **Attach** each part to the **acid-free card** with tiny dots of archival glue or thin strips of archival tape. Never use regular tape.
4. **Label each part in pencil** beside it: sepal, petal, stamen (anther + filament), stigma, style, ovary slice, leaf.
5. **Fill in the specimen label** in the bottom-right corner of your card, using the fields below.

LAYOUT PLAN

specimen label

SPECIMEN LABEL: COPY THESE FIELDS ONTO YOUR CARD

Specimen	Lily (<i>Lilium</i> sp.) Variety: _____
Parts mounted	
Source	(florist, grocery store, garden...)
Collected and pressed by	
Date pressed / Date mounted	/

6. Regular paper contains acid. Over many years it turns yellow and crumbly. Explain why scientists use acid-free paper for specimens they want to study 100 years from now.

7. A bee visits this lily, then another lily. Trace one pollen grain from where it is made to where a seed will form. Use: **anther, stigma, style, ovary, ovule**. Mark where **pollination** ends and **fertilization** begins.

Wrap-up**What the Lily Taught You**

Answer these with your mounted specimen in front of you. First, go back to the **From memory** table on page 2 and fill in the last column.

8. Which part on your card is best at making glucose? Most of the flower isn't green, so where did the lily get the energy to build it? Use **photosynthesis** in your answer.

⚠ THINK IT THROUGH

A classmate says: "*Lilies are big and colourful so that people will buy them.*"

9. Look back at your prediction on page 1. Agree or disagree with your classmate, using **two pieces of evidence** from your dissection (think: colour, spots, smell, the groove, where the anthers stick out).

★ Florist lilies have been **selectively bred** by people for many years. Name one trait people probably made bigger or brighter, and one trait that was there first for pollinators.

Me How Did I Do?

I can...	Yes	Mostly	Not yet
find the sepals, petals, stamens, and pistil on a real lily	☐	☐	☐
explain the job of the stamen and the pistil in making seeds	☐	☐	☐
explain the difference between pollination and fertilization	☐	☐	☐
press and mount a specimen so it will last	☐	☐	☐

One flower part I could now explain to a Grade 5 student: _____